

Cancer cell invasion driven by extracellular matrix remodeling is dependent on the properties of cancer-associated fibroblasts

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Abstract

Purpose As one form of tumor invasion, cancer cells can invade the extracellular matrix (ECM) through tracks that have been physically remodeled by cancer-associated fibroblasts (CAFs). However, CAFs are a heterogeneous population with diverse matrix-remodeling capacities. The purpose of this study was to investigate how CAFs with various matrix-remodeling capacities influence cancer cell invasion.

Methods We established single-cell-derived clones from three primary cultures of CAFs from lung adenocarcinoma patients (Case 1, 5 clones; Case 2, 5 clones; and Case 3, 7 clones). Using a co-culture model, we evaluated the correlations between the number of invaded cancer cells and the remodeling areas generated by CAF clones in each case.

Results When A549 lung adenocarcinoma cells and CAF clones were co-cultured, both the numbers of invaded cancer cells and the remodeling areas

generated by the CAF clones varied greatly. The number of invaded cancer cells was moderately and strongly correlated with the remodeling areas generated by each CAF clone originating from Cases 1 and 2 (R^2 value = 0.53 and 0.68, respectively), suggesting that the remodeling areas in the ECM may determine the number of invaded cancer cells. In contrast, the number of invaded cancer cells was not correlated with the remodeling areas generated by CAF clones originating from Case 3, suggesting that factors other than the remodeling areas might determine the number of invading cancer cells.

Conclusions These findings showed two types of fibroblast-dependent cancer cell invasion that are dependent on and independent of the remodeling areas generated by CAFs.

Keywords Cancer-associated fibroblasts · Fibroblast-dependent cancer invasion · Extracellular matrix remodeling · Single-cell-derived clones

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Introduction

Cancer tissue is composed of cancer cells and stromal cells, including immune cells, endothelial cells, and fibroblasts (Bissell and Radisky 2001; Kalluri and Zeisberg 2006; Tlsty and Coussens 2006). These stromal components organize the microenvironment for cancer cells and influence the proliferation and invasion of cancer cells (Allinen et al. 2004; Kharaishvili et al. 2014; Orimo et al. 2005; Paulsson and Mücke 2014; Pietras et al. 2008). Cancer-associated fibroblasts (CAFs) communicate with cancer cells and play important roles in cancer cell invasion through the over-secretion of growth factors, such as transforming growth factor β (TGF β) 1, chemokine (C-X-C motif) ligand 12 (CXCL12), and matrix metalloproteinases (Allen and Louise Jones 2011; Cameron et al. 2000; Jodele et al. 2006; Kalluri and Zeisberg 2006; Kharaishvili et al. 2014; Orimo et al. 2005; Paulsson and Mücke 2014). Moreover, CAFs facilitate the local invasion of cancer cells by biomechanically remodeling the extracellular matrix (ECM), and cancer cells invade by following the CAFs into the matrix (Fig. 1a) (Friedl et al. 2012; Gaggioli et al. 2007; Neri et al. 2015; Satoyoshi et al. 2015). This type of cancer cell invasion is not dependent on the properties of cancer cells, such as proteases and Rho function (Gaggioli et al. 2007). Instead, the Rho/ROCK signaling pathway in CAFs is necessary for ECM remodeling, which physically contributes to cancer cell invasion (Gaggioli et al. 2007). We recently reported that podoplanin expressed in CAFs increases their RhoA activity, and increases in the extents of the remodeling areas induced by podoplanin-expressing CAFs favor the local invasion of cancer cells (Ito et al. 2012; Neri et al. 2015). Considering these results, the extents of the CAF-induced remodeling areas in the ECM might determine cancer cell invasion. Meanwhile, the deposition of matrix components, such as fibronectin and tenascin C, has been observed in remodeling areas generated by CAFs (Gaggioli et al. 2007). As fibronectin and tenascin C are well-known matrix components that promote cancer cell invasion (Allen and Louise Jones 2011; Guttery et al. 2010), biochemical factors in the remodeling areas created by CAFs might also be important for cancer cell invasion, in addition to biomechanical factors such as the extents of the remodeling areas.

To address this issue, the effects of a variety of biomechanical and biochemical characteristics of CAFs on ECM remodeling should be investigated. In this study, we established single-cell-derived clones from primary cultures of CAFs originating from three different lung adenocarcinoma patients. These CAF clones created various extents of remodeling areas in the ECM. We then studied the correlations between the remodeling areas generated by the CAF clones

and the numbers of invaded cancer cells using an in vitro co-cultured system.

Materials and methods

Cell culture

CAFs were obtained from the surgically resected tumors of lung adenocarcinoma patients and cultured as described before (Hoshino et al. 2008; Ishii et al. 2010; Neri et al. 2015). All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. All the specimens were collected after the subjects had provided their written informed consent, and the specimen collection was approved by the institutional review board of our institution. The human lung adenocarcinoma cell line, A549 (American Type Culture Collection, Manassas, VA), was cultured in Dulbecco's modified Eagle's medium/Nutrient Mixture F-12 Ham (DMEM/F-12, Sigma-Aldrich, St. Louis, MO) containing 10 % FBS and 1 % penicillin and streptomycin. All the CAFs and cell line were incubated at 37 °C in an atmosphere containing 5 % CO₂.

Expression of DsRed2 fluorescent proteins in cancer cells

A549 cancer cells were transfected with the pDsRed2-Nuc vector (Clontech Laboratories, Mountain View, CA) using LipofectAMINE 2000 reagent (Life Technologies, Carlsbad, CA) according to the manufacturer's instructions. Stable transformants were selected using G418 (Sigma-Aldrich) and FACSaria II cell sorting (BD Biosciences, Franklin Lakes, NJ). The localization of pDsRed2-Nuc in the nucleus of cancer cells facilitated the measurement of the number of cancer cells for the in vitro experiments.

Quantitative real-time reverse transcriptase-polymerase chain reaction (qRT-PCR)

Cells were suspended in 1 mL of TRIzol (Life Technologies) and stored at −80 °C. Total RNA was purified from thawed samples using standard techniques, and cDNA was synthesized using the PrimeScript RT reagent Kit (Takara Bio, Shiga, Japan), according to the manufacturer's instructions. qRT-PCR was performed in a Smart Cycler System (Takara Bio) with SYBR Premix Ex Taq II (Takara Bio) according to the manufacturer's instructions. The primers we used are given in Supplementary Table 1.

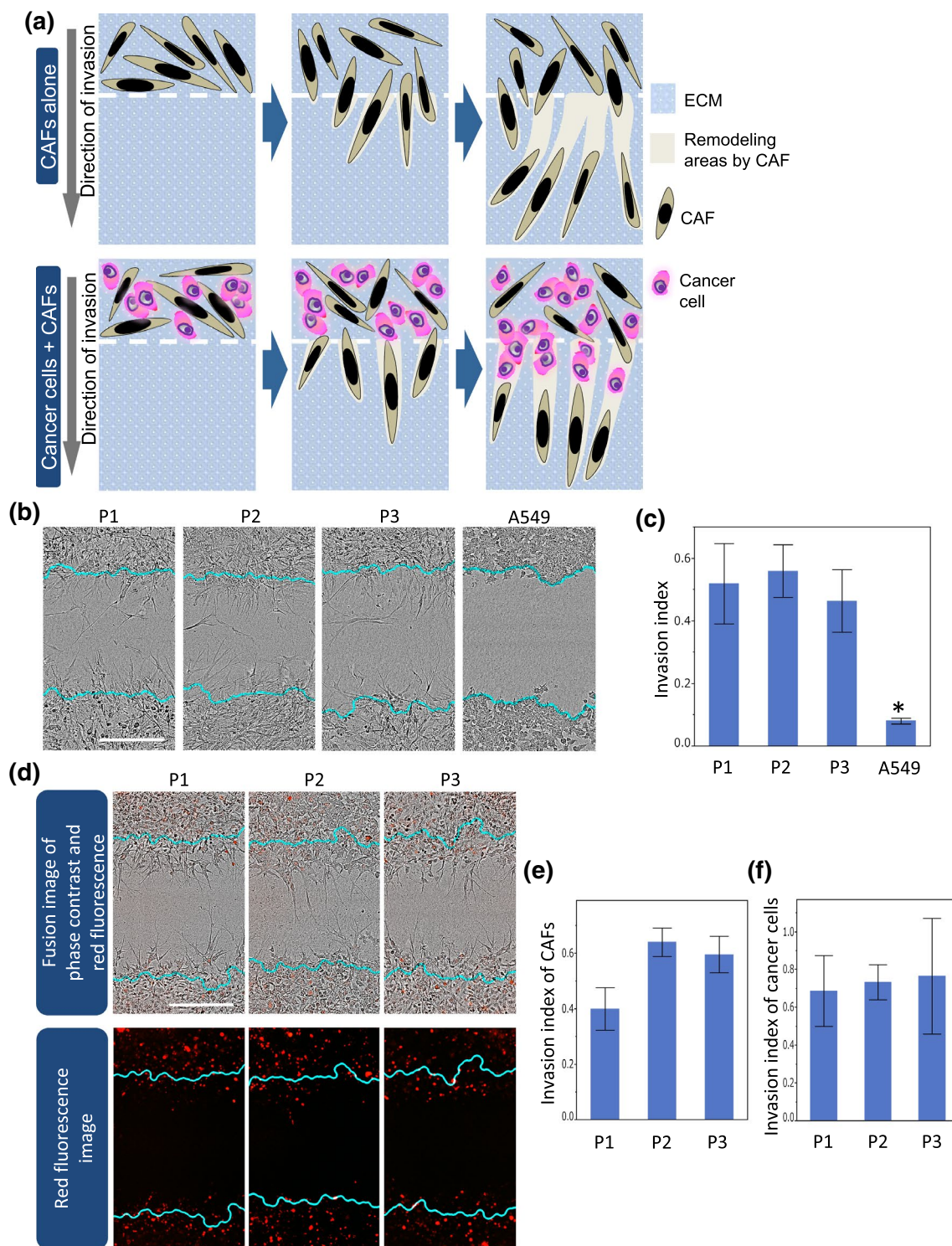


Fig. 1 Invasion abilities of three primary cultures of CAFs and co-cultured cancer cells. **a** Scheme showing the form of invasion in the ECM for CAFs alone and co-cultures of CAFs and cancer cells. Cancer cells invade by following the CAFs, indicating a fibroblast-dependent form of cancer cell invasion. Examples **(b)** and quantification **(c)** of the invasion of the three parental CAFs and A549 cancer

cells cultured alone for 48 h using the collagen invasion assay model. Examples **(d)** and quantification **(e, f)** of the invasion of co-cultures of A549 cancer cells and parental CAFs for 48 h using the collagen invasion assay model. P1, P2, and P3 represent the parental CAFs from Cases 1, 2, and 3, respectively. Values are the mean $\pm$ SEM. Bar 300 μ m. * $P < 0.001$

Lentiviral vectors

Lentivirus vectors containing human telomerase reverse transcriptase (hTERT) were generated by subcloning the pBabe-Puro-hTERT vector (Counter et al. 1998; Masutomi et al. 2003) into the CSII-CMV-RfA-IRES2-Venus vector (RIKEN BioResource Center, Ibaraki, Japan). First, cDNA for hTERT was inserted into pENTR2B (Life Technologies). Thereafter, LR recombination reaction was performed between the entry clones and CSII-CMV-RfA-IRES2-Venus (RIKEN BioResource Center), using Gateway LR Clonase (Life Technologies) according to the manufacturer's instruction. The lentiviruses were produced using 293T cells transfected with PCAG-HIV, pCMV-VSV-G-RSV-Rev, and hTERT vectors. Infection was achieved using the LipofectAMINE 2000 reagent, according to the manufacturer's instructions. Vector-containing medium was filtered through a 0.45- μ m filter, and 8 μ g/mL of polybrene (Santa Cruz Biotechnology, Dallas, TX) was added for target cell transduction.

Collagen invasion assay

According to the manufacturer's instructions, 0.3 % type I collagen gel was coated on 96-well plates (Essen ImageLock; Essen BioScience). Cells were then plated in the collagen-coated 96-well plates at a density of 5×10^4 cells/well (co-culture of CAFs and cancer cells; 2.5×10^4 cells each). After 1 h of incubation, a wound was made in the cell layer using the 96-well WoundMaker (Essen BioScience), and the cells were embedded in type I collagen gel. Fresh medium was added, and scratched field images were obtained using IncuCyte Live-Cell Imaging System. Time-lapse fluorescence and phase-contrast images were obtained at 3-h intervals for 48 h. Each data element was measured in at least eight fields (magnification $\times 10$) in two or more independent experiments.

In this collagen invasion assay, we observed that CAFs invaded the collagen matrix to a great extent and that cancer cells followed the leading CAFs and invaded within the tracks created by the CAFs, suggesting fibroblast-dependent cancer cell invasion (Fig. 1a) (Gaggioli et al. 2007; Neri et al. 2015).

Establishment of single-cell-derived clones from primary cultures of CAFs

Using the FACSaria II cell sorter, CAFs expressing hTERT-Venus were sorted one by one into each well of 384-well plates. Then, fluorescence images were obtained every 12 h using the IncuCyte Live-Cell Imaging System (Essen BioScience, Ann Arbor, MI). When an increase in the fluorescent intensity of a well was observed, we

retrogradely confirmed whether a single cell had been present in the well immediately after the sorting process. We established CAF clones from single cells of CAFs. When single-cell-derived clones reached semi-confluence after about 2–3 weeks of incubation, they were replated on 96-well plates. Similarly, when they reached semi-confluence, they were plated on 24-well, 12-well, and 6-well plates and in 6- or 10-cm dishes, sequentially (Fig. 2a, b; Supplementary Movie 1). Finally, we obtained a total of 17 single-cell-derived clones from three primary cultures of CAFs. In Cases 1 and 2, the pathological stages were early (Stage IA and IB, respectively), whereas the pathological stage of Case 3 was advanced (Stage IIIA) (Supplementary Table 2).

Image analysis

The image analysis was performed using CL-Quant software (Nikon, Tokyo, Japan). Algorithms to detect the measurement objects were developed for each experiment as follows (Supplementary Fig. 1). (1) The cell region was detected using the first frame of the phase-contrast images. (2) The invasion region was determined by subtracting the cell region from the whole image region. (3) The invaded CAF region within the invasion region was detected using all the time frames of the phase-contrast images. (4) The remodeling area generated by the CAFs was determined by overlaying the invaded CAF region on the previous frame of the remodeling area for all the time frames. (5) The number of cancer cells in the invasion region was counted using red fluorescence images for all the time frames. (6) The invasion index for the CAFs was determined as the average of the remodeling area generated by the CAFs/the invasion region. (7) The invasion index for the cancer cells was determined as the average $\times 1000$ of the number of invaded cancer cells/the invasion region.

Statistical analysis

Statistical analysis was performed using JMP 9 (SAS Institute, Cary, NC). Statistical significance was assessed by Student's *t* test. The coefficient of determination (R^2 value) was calculated. *P* values of <0.05 were considered statistically significant.

Results

Invasion of cells from three primary cultures of CAFs and co-cultured cancer cells

First, we investigated the invasion ability of cells from three primary cultures of CAFs and A549 cancer cells.

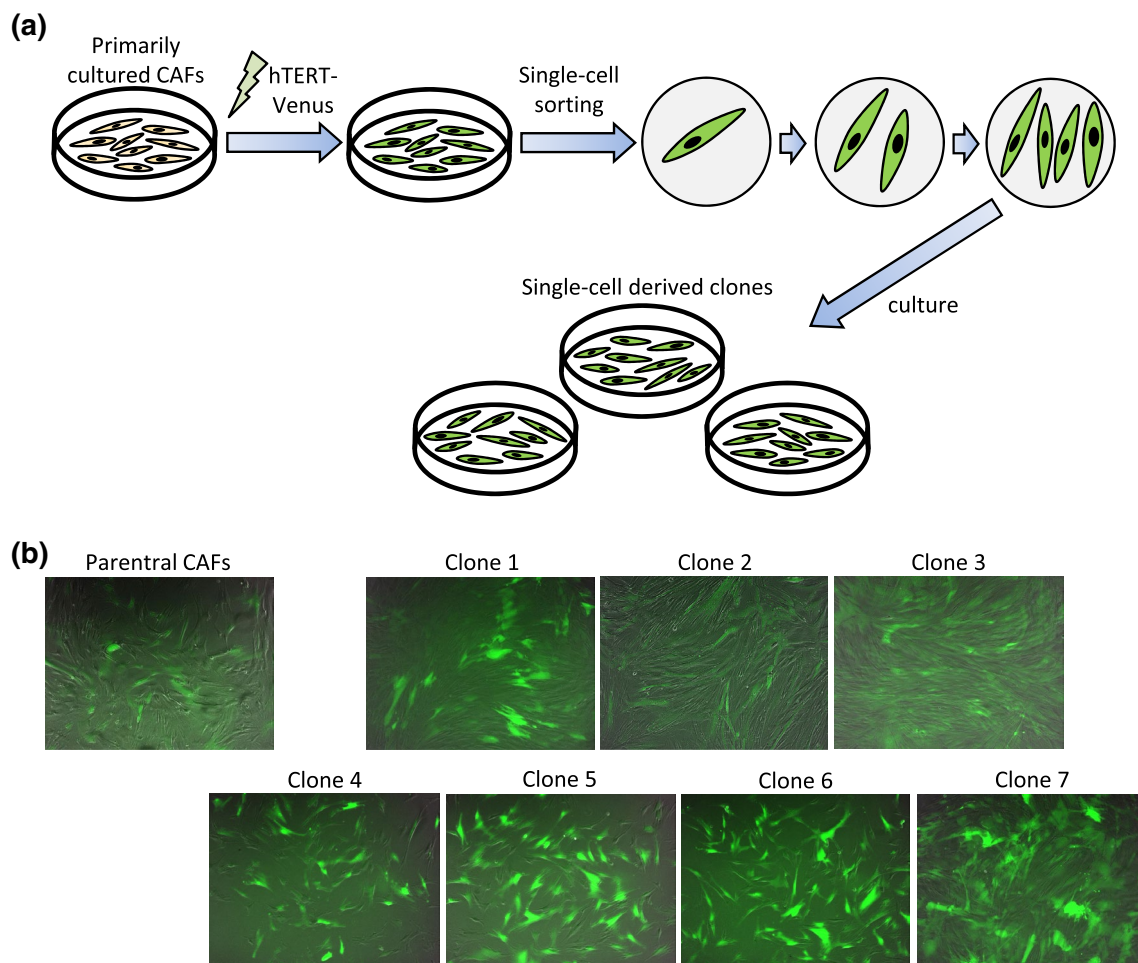


Fig. 2 Establishment of single-cell-derived clones from primarily cultured CAFs infected with hTERT. **a** Scheme showing the method used to establish single-cell-derived clones from the primary cultures of CAFs (see “Materials and Methods”). **b** Examples of the paren-

tal CAFs from Case 3 and their single-cell-derived clones. hTERT-infected CAFs express Venus in their cytoplasm (green). The images were fusion images of phase contrast and green fluorescence

We found no differences in the extents of the remodeling areas created by the three parental CAFs, and cells from the three primary cultures of parental CAFs showed a significantly greater extent of invasion than the A549 cancer cells (Fig. 1b, c). Similarly, when the CAFs and cancer cells were co-cultured, no significant differences were found in the extents of the remodeling areas generated by the CAFs and the numbers of invaded co-cultured cancer cells (Fig. 1d–f).

Establishment of single-cell-derived clones from primary cultures of CAFs infected with hTERT

We established a total of 17 single-cell-derived clones from three primary cultures of CAFs originating from lung adenocarcinoma patients (Case 1, 5 clones; Case 2, 5 clones; and Case 3, 7 clones) using hTERT transformation of the CAFs (Fig. 2a, b; Supplementary Movie 1).

Invasion of CAF clones and co-cultured cancer cells

Next, we analyzed the invasion abilities of CAFs among the clones from each primary culture of CAFs. When the CAF clones were cultured alone in an in vitro invasion assay, a diverse range of extents of remodeling areas was observed (Fig. 3a, b). The median values of the CAF invasion index were 0.416 (0.329–0.551), 0.803 (0.408–0.847), and 0.549 (0.252–0.674) for Cases 1, 2, and 3, respectively. Similarly, when CAF clones were co-cultured with cancer cells, the extents of the remodeling areas generated by the CAF clones and the numbers of invaded co-cultured cancer cells varied greatly (Fig. 4a–c). The median values of the CAF invasion index were 0.404 (0.312–0.431), 0.575 (0.325–0.657), and 0.500 (0.361–0.614) for Cases 1, 2, and 3, respectively. The median values of the invasion index of cancer cells co-cultured with the CAF clones were 0.701 (0.283–1.06), 0.659 (0.408–0.864), and 0.672 (0.435–0.767) for Cases 1, 2, and 3, respectively.

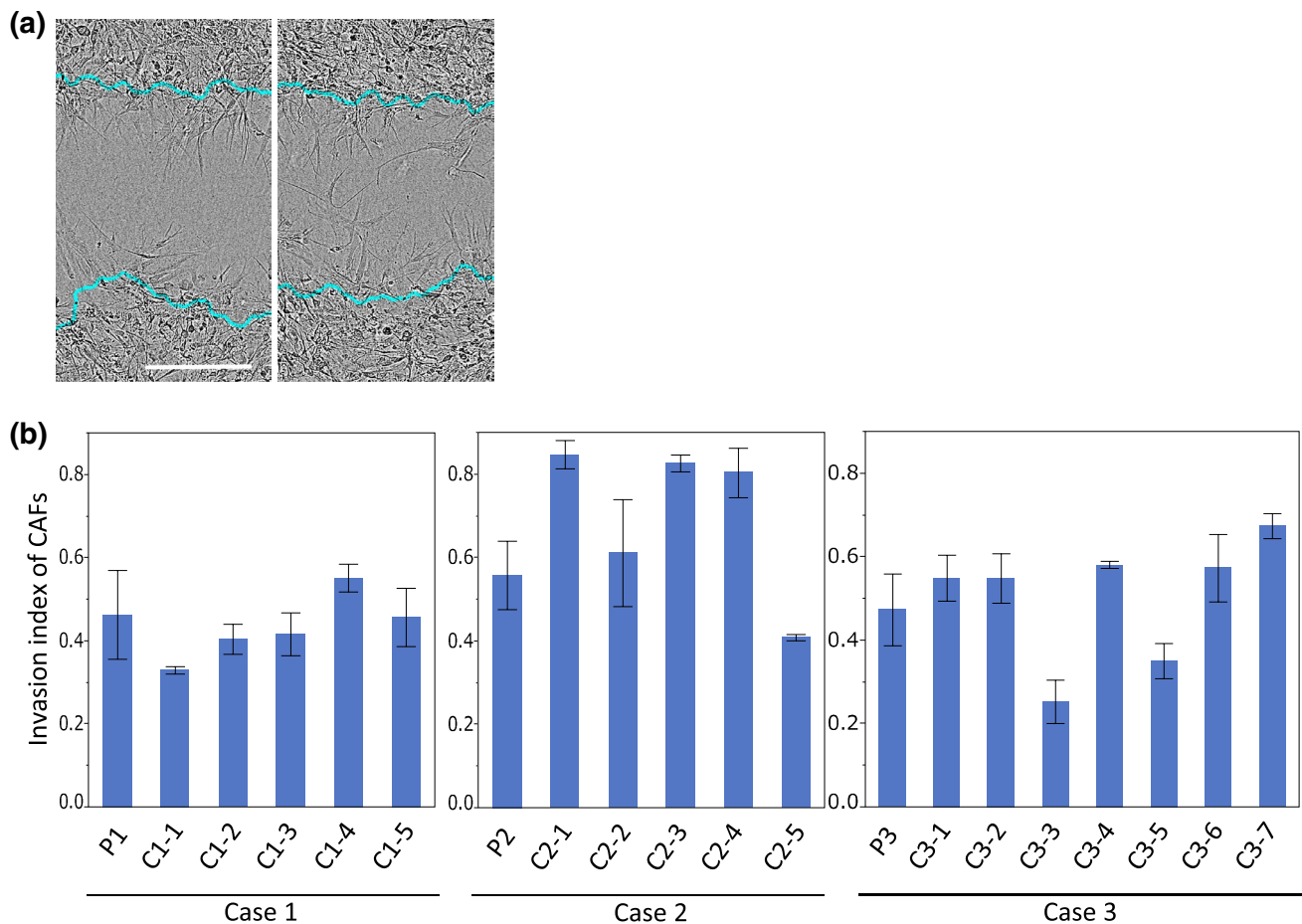


Fig. 3 Invasion abilities of CAF clones cultured alone in collagen gel. Representative examples (a) and quantification (b) of the invasion of CAF clones cultured alone for 48 h using the collagen inva-

sion assay model. P1, P2, and P3 represent the parental CAFs of Cases 1, 2, and 3, respectively. Values are the mean $\pm$ SEM. Bar 300 μ m

Relationship between the invasion of CAF clones and the invaded cancer cells

We investigated the correlations between the extents of the remodeling areas created by the CAFs and the number of invaded cancer cells (Fig. 5a). A moderate correlation was present between the number of invaded cancer cells and the extents of the remodeling areas generated by the CAF clones in Case 1 ($R^2 = 0.53$). In Case 2, a strong correlation was observed between the number of invaded cancer cells and the extents of the remodeling areas generated by the CAF clones ($R^2 = 0.68$). On the other hand, the number of invaded cancer cells was not correlated with the extents of the remodeling areas generated by CAF clones in Case 3 ($R^2 = 0.063$).

Discussion

CAFs are known to form a heterogeneous population with diverse biological properties (Bauer et al. 2010; De Wever

et al. 2008; Paulsson and Micke 2014). Taking advantage of this property, we established single-cell-derived clones with different invasion abilities and analyzed the influence of these cells on cancer cell invasion using co-cultures. This is the first report to describe specifically the establishment and investigation of single-cell-derived clones of CAFs.

First, we analyzed the invasion of parental CAFs and cancer cells using an in vitro collagen invasion model. No significant differences were observed among them. Subsequently, among the single-cell-derived clones from each parental CAF, the invasion abilities of the CAF clones and the co-cultured cancer cells varied greatly. These findings provided evidence of the biological heterogeneity of CAFs within the same tumor, suggesting that analysis at the single-cell level could emerge as a powerful tool for investigating the heterogeneous properties of CAFs.

Some studies have previously reported that the form of invasion used by cancer cells depends on biomechanical remodeling of the ECM by CAFs (Gaggioli et al. 2007; Neri et al. 2015; Satoyoshi et al. 2015) and that the

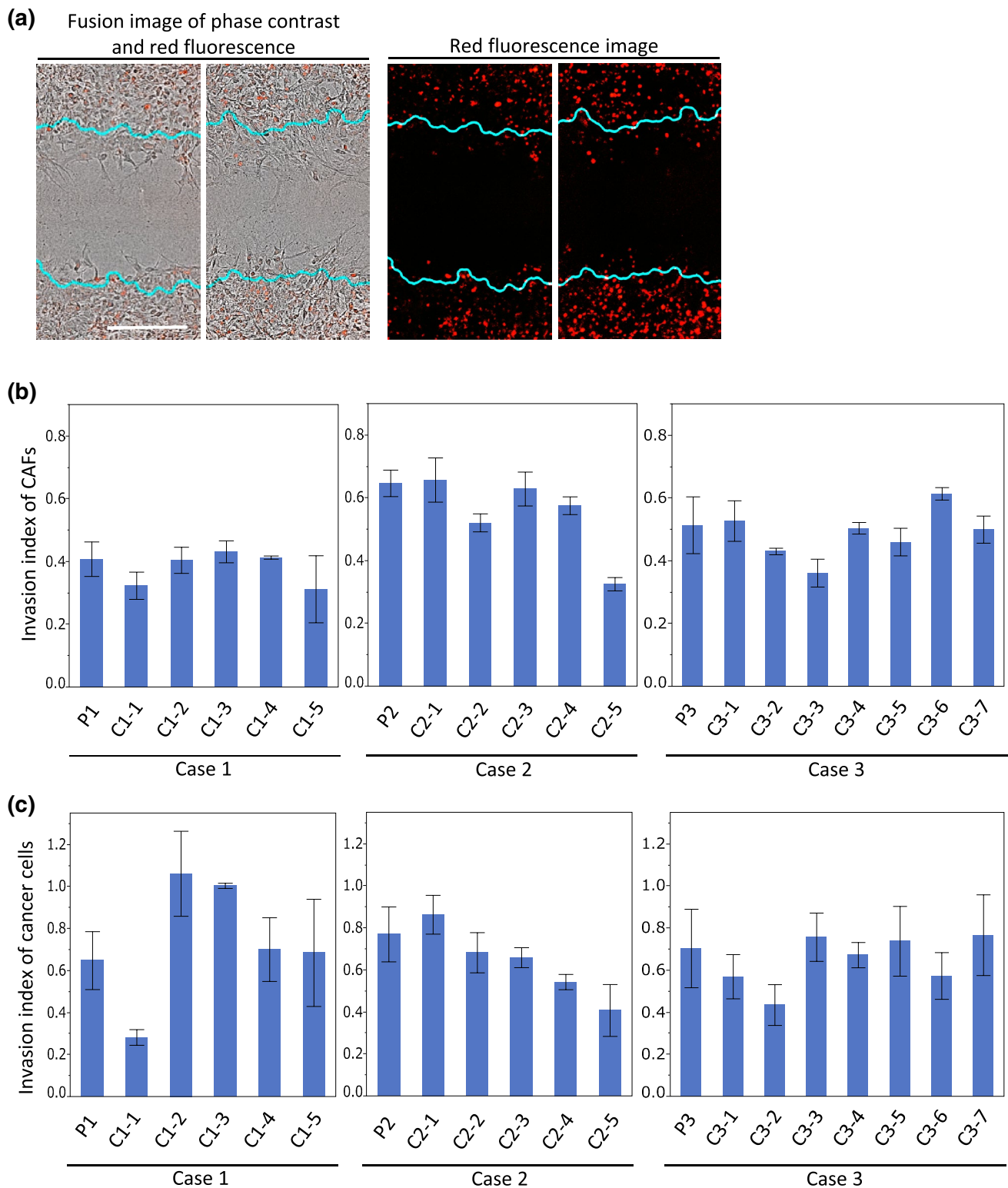


Fig. 4 Invasion abilities of cancer cells and CAF clones cultured together in collagen gel. Representative examples (a) and quantification (b, c) of the invasion of co-cultures of A549 cancer cells and

CAF clones for 48 h using the collagen invasion assay model. P1, P2, and P3 represent the parental CAFs of Cases 1, 2, and 3, respectively. Values are the mean $\pm$ SEM. Bar 300 μ m

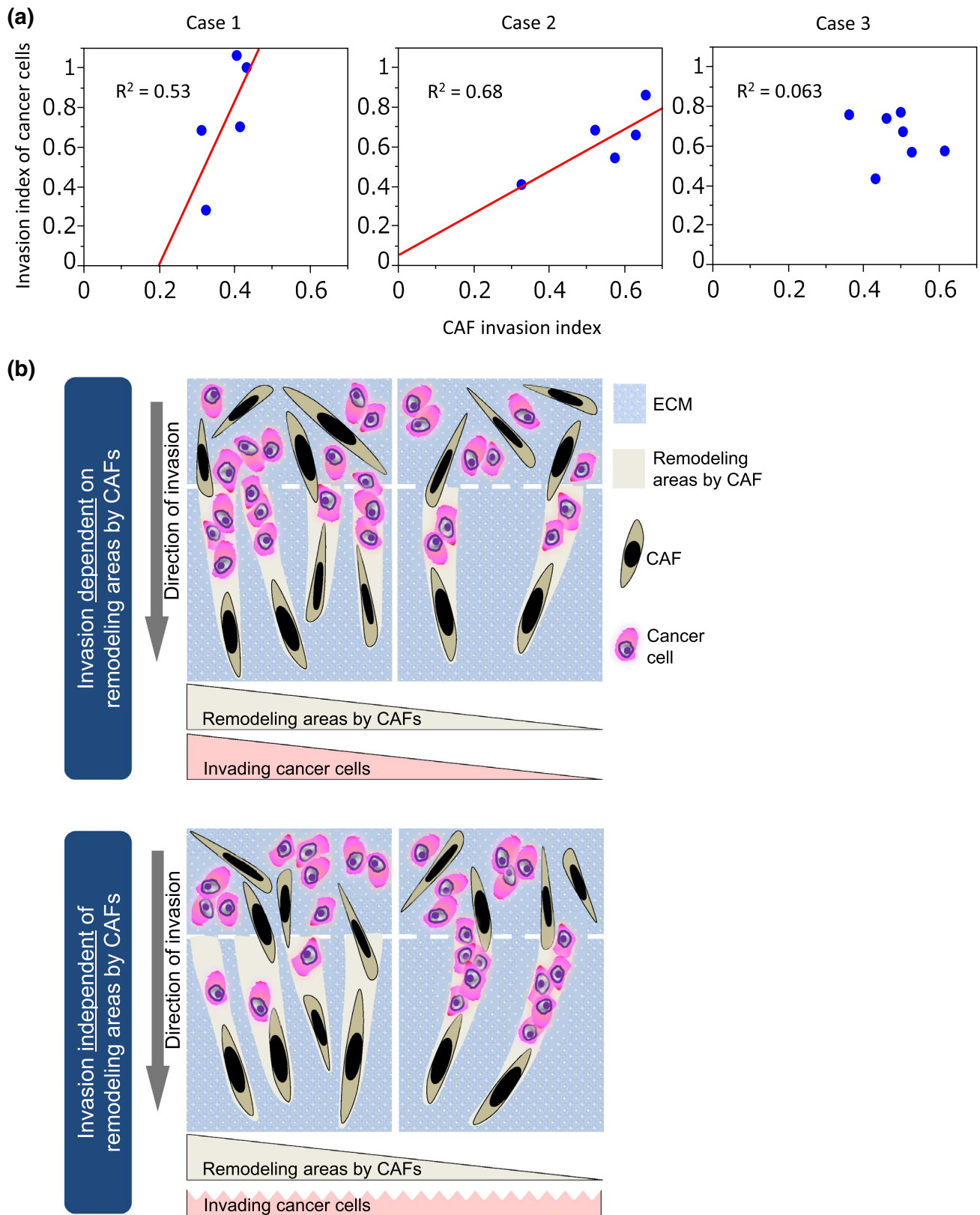


Fig. 5 **a** Relationship between the invasion of CAFs and the invaded cancer cells in Cases 1, 2, and 3. R^2 = coefficient of determination. **b** Scheme showing two types of cancer cell invasion, one dependent on and one independent of the remodeling areas generated by CAFs

biochemical attachment of matrix components, such as fibronectin and tenascin C, can be found in the areas that have been remodeled by CAFs (Gaggioli et al. 2007). However, whether the extents of the remodeling areas or certain biochemical factors in the remodeling areas generated by CAFs are important for cancer cell invasion remains unclear. In Cases 1 and 2 of this study, positive correlations between the number of invaded cancer cells and the extents of the remodeling areas generated by CAF clones were observed. The local invasion of cancer cells in Cases 1 and 2 was thought to depend on the extents of the remodeling areas generated by each CAF clone (Fig. 5b, upper). On the other hand, cancer cell invasion was not correlated with the extents of the remodeling areas of CAF clones in Case 3. Although the cancer cells invaded by following the CAFs into the matrix in all three cases, the cancer cell invasion in Case 3 might have depended on not the extents of the remodeling areas, but on biochemical factors (Fig. 5b, lower). The ability of each CAF clone in Case 3 to facilitate cancer cell invasion might have differed for the following possible reasons: (1) the CAF clones might have secreted different levels of growth or matrix factors that enhanced the migration of the cancer cells (Bremnes et al. 2011; Gaggioli et al. 2007; Kalluri and Zeisberg 2006; Ohlund et al. 2014), or (2) the CAFs might have interacted with the cancer cells to induce a change in the invasion ability of the cancer cells (Bonaventure et al. 2013; Taddei et al. 2013). As for possibility 1, we measured the relative mRNA expressions of hepatocyte growth factor (HGF), TGF β 1, CXCL12, fibronectin, and tenascin C in the CAF clones, but no significant positive correlations between the expression level of each mRNA and the number of invaded cancer cells were observed (Supplementary Fig. 2). In growth factors, there were no correlations between the number of invaded cancer cells and the expression level of HGF, TGF β 1, and CXCL12. In ECM factors, while there was no correlation in the expression level of fibronectin, the negative correlation between the number of invaded cancer cells and the expression level of tenascin C was found. We need to investigate whether tenascin C in CAFs suppresses the local invasion of cancer cells, although it is reported that tenascin C promotes tumor growth and metastasis (Allen and Louise Jones 2011; Oskarsson et al. 2011). These results suggested that the ability of CAFs to secrete other biochemical factors or the combination of some of the factors might have enhanced cancer cell invasion. These possibilities suggest that the number of invading cancer cells in the remodeling areas generated by CAFs may depend on various biochemical and biomechanical factors of CAFs on a case-by-case basis.

The CAFs from Cases 1 and 2 were from early-stage cancer (Stage I), whereas the CAFs from Case 3 were from an advanced-stage cancer (Stage III). During the early

stage of cancer (Cases 1 and 2), the CAFs may have heterogeneous abilities to induce ECM remodeling and relatively homogeneous abilities to secrete growth and matrix factors to facilitate cancer cell invasion; thus, the number of invaded cancer cells was correlated with the extents of the remodeling areas generated by the CAF clones as a biomechanical factor. At an advanced stage of cancer (Case 3), however, the biochemical factors associated with ECM remodeling and degradation may differ among the CAF clones, as well as the extents of the remodeling areas. Further investigation of the characteristics of ECM remodeling and ECM degradative enzyme expression induced by CAFs according to the stage of cancer may be useful.

In this study, we reported the establishment of single-cell-derived clones of CAFs and found that positive correlations between the number of invaded cancer cells and the extents of the remodeling areas of CAF clones were observed in some cases, whereas cancer cell invasion was not correlated with the extents of the remodeling areas of CAF clones in the other case, suggesting that biomechanical and biochemical features of ECM remodeling generated by CAFs might be rate-limiting factors for fibroblast-dependent cancer cell invasion. Our findings also suggest that further investigation of how CAFs with various phenotypes influence the local invasion of cancer cells could be helpful for understanding the mechanisms of heterogeneous forms of cancer cell invasion in intratumoral microenvironments.

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Compliance with Ethical Standards

Conflict of interest The authors have no conflicts of interest to disclose.

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Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

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